Many-target crossover: amortised compact-orbit IC against batch rho (2026-10-30)

Verdict: dead where it can be ranked ($n = 41..61$); undecidable at $n = 73, 83$ only for want of counts, and the $n = 83$ wall-clock evidence points the same way. Amortising relation collection over k targets never makes IC cheaper than batch rho. With $G = \min_k \frac{\mathrm{IC}(k)}{\text{batch rho}(k)}$: clock-priced `ic price` units give $G = 3.0$ to 4.1 at $n = 41..61$ (best swept recipe); no size has $G < 1$; G is flat to rising above 2^{36}. The §5a count band (IC at its emitted additions, a lower bound; rho at 0, 0.1, 1 GAE per canonicalisation) keeps $G > 1$ at $n = 41, 43, 53, 61$ (3.1 to 4.8 at $w = 0$, 1.6 to 2.4 at $w = 1$); at $n \le 45$ and $n = 47$ IC's descent and construction counts are not emitted, so those sizes are unresolved by counts. $n = 83$ (wall seconds, practicality only): $G = 4.8$ to 5.1, and a precomputation-rho table with the same setup time has a smaller per-target cost than IC at every k. Class: accounting. Not an improvement to the IC baseline.

Model. $\mathrm{IC}(k) = S_0 + kd$; batch $\mathrm{rho}(k) = \rho_1 K(k)$ with $K(k) = \sum_{i<k} \binom{2i}{i} 4^{-i} \approx 2\sqrt{\frac{k}{\pi}}$ (Kuhn–Struik law as implemented in `src/bin/ic/rho.rs`, $\frac{K(32)}{32} = 0.19869$). A window exists iff $\sigma\delta < 0.318$ with $\sigma = \frac{S_0}{\rho_1}$, $\delta = \frac{d}{\rho_1}$; $G \approx 1.773\sqrt{\sigma\delta}$. The product $S_0 d$ must fall by G^2 (9 to 17x at $n = 41..61$, about 23x at $n = 83$).

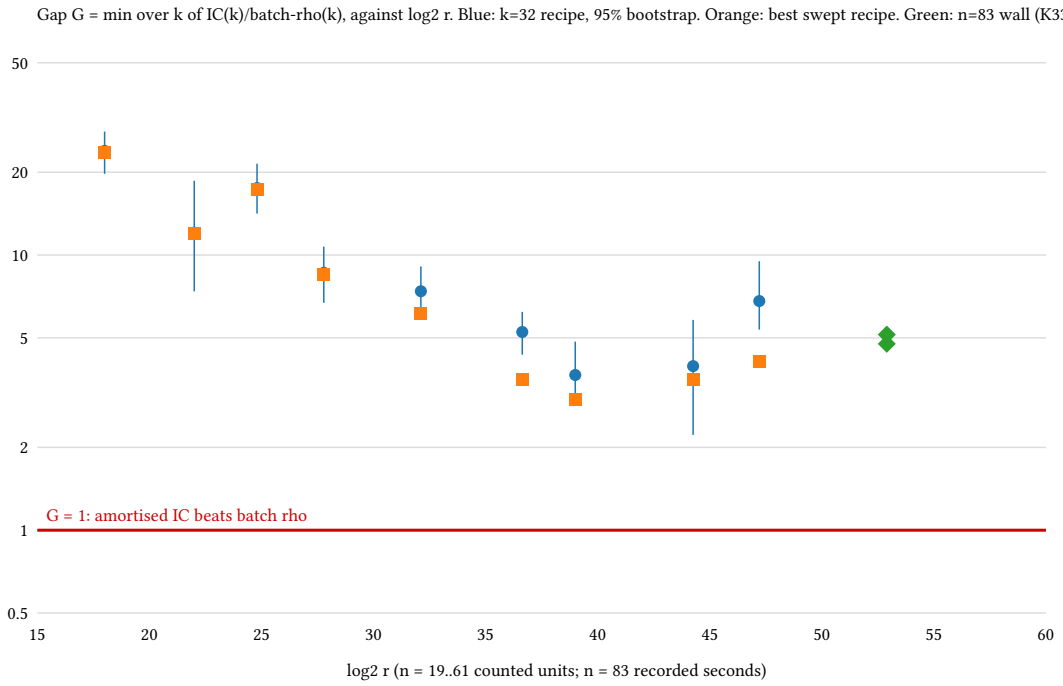


Figure 1: Gap G against $\log_2 r$. Blue: k=32 recipe with 95% bootstrap; orange: best swept recipe; green: n=83 wall (K3300). The red line $G = 1$ is where amortised IC would win.

curve	log2 r	S0/rho1	d/rho1	G [95%]	k_opt	G best recipe
K1/2^19	18.0	95.1	1.903	23.90 [19.74, 28.10]	50	23.64
K1/2^23	22.0	51.0	0.904	12.06 [7.38, 18.62]	57	12.00
K1/2^45	24.8	124.5	0.784	17.52 [14.15, 21.48]	159	17.35
K0/2^37	27.8	40.1	0.587	8.62 [6.70, 10.73]	69	8.50
K1/2^43	32.1	36.3	0.477	7.39 [5.91, 9.09]	76	6.11
K1/2^47	36.6	33.6	0.261	5.25 [4.34, 6.21]	129	3.53
K0/2^41	39.0	25.1	0.170	3.66 [2.84, 4.85]	148	2.98
K0/2^53	44.3	55.7	0.089	3.95 [2.22, 5.81]	626	3.52
K0/2^61	47.2	80.8	0.182	6.80 [5.36, 9.49]	444	4.10

Measured vs extrapolated. Measured: S_0, d, ρ_1 for $n = 19..61$ (clock-priced `ic price` units and emitted counts, same process, 4 target sets x 9 sizes plus the recipe sweep); $n = 83$ S_0, d (16 targets), ρ_1 (16 plain-rho runs) in CPU seconds on one runner per row. Ledger only: $n = 73$ (one target, ρ_1 scaled from $n = 83$): $G = 15$ to 22 for the untuned K600 recipe. Extrapolated: the trend fit (slope -0.054 per bit, $G(73) = 2.3[1.6, 3.0]$, $G(83) = 2.7[2.0, 3.5]$), which is not relied on: the top four sizes are flat to rising.

n = 83, total CPU seconds for k targets (run 2, K3300). IC 825 + 0.049 k; BL table (a): 651 + 0.0042 k with a 0.3 GB table; batch rho 2.375 K(k): batch rho is cheapest at every k from 1 to 10^7 and BL(a) dominates IC at every k.

Not run: the equal-advice instrument (needs 24 to 48 GiB, 40 to 100 core-hours); counted S_0 at $n = \frac{73}{83}$ is not emitted by the producer. See README.md for the model, uncertainty method, section 5a band, literature check (one gap: Kuhn–Struik text unreadable here), and limits.